

What Does This Test Support?

Unit 4 · Topic 4.10 · THE PROBLEM

Suppose this term 4,000 students chose Focus mode and 3,600 chose Guided. Independent random samples of 100 students from each mode completed the same practice set. The service asked in advance whether mean Focus time was greater and set $\alpha = 0.05$. Only a difference of at least 3 minutes matters in practice.

Nia: “The $p \approx 0.0263 < 0.05$ supports a higher Focus mean. So Guided causes every student to finish at least 3 minutes faster; switch everyone.”

Mateo: “The sample gap is 1.2 minutes, and students chose their modes. A 3-minute effect and causation are not established. So retain H_0 ; there is no evidence of different population means.”

Practice time summaries (minutes)

Mode	N	n	$\bar{x}$	s
Focus	4,000	100	42.8	4.2
Guided	3,600	100	41.6	4.5

FIRST TAKE — one sentence before you work the parts

Which claim seems too strong: no average difference, or Guided helps everyone? Give one reason.

- DOK 1** (a) State H_0 and H_a for the population mean times in Focus-minus-Guided order.
- DOK 2** (b) The difference is 1.2 minutes and $SE = 0.6155485$. Calculate t and interpret the supplied one-sided $p = 0.0263$ in one sentence.
- DOK 3** (c) In 4 sentences, combine the supported parts of both reports. Use the p-value for the mean claim, compare 1.2 with the 3-minute benchmark, and explain why self-selection and a mean difference do not justify switching every student.

Self-paced bonus problem. Work (a)–(c) from this sheet; turn it in whenever you finish.